

SAFDAR NIZAM MOHAMMED

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Work Experience

Help Us Grow Foundation

AI Systems Engineer

Florida, USA

May 2025 to November 2025

- Built an internal retrieval platform using LangChain, Flowise, Llama 3.1, and Mistral with routing logic, metadata indexing, vector search optimization, and quality controls which increased retrieval consistency and reduced query latency. Added evaluation workflows for RAG systems which improved output reliability by 20 percent.
- Developed backend services with Python, FastAPI, and Pydantic with schema refinement, automated health checks, observability hooks, and scheduled indexing processes which supported seven thousand monthly users and reduced backend error rates by 22 percent.
- Designed event driven workflows using Google Sheets API, Monday dot com, Zapier, and n8n with validation rules, synchronization logic, and automated data integration which reduced manual update time by ten hours per week.

J and G Software Solutions

AI Engineer

Florida, USA

October 2024 to April 2025

- Improved a React, Tailwind CSS, and TypeScript front end with clearer component boundaries, optimized render flow, accessibility adjustments, and routing improvements which increased interaction performance by 14 percent.
- Enhanced REST and GraphQL layers with authentication checks, cache refinement, error monitoring, and structured failure reporting which reduced API failure frequency by a factor of one point two five.
- Added CI based schema validation and automated contract testing for REST and GraphQL endpoints using TypeScript tooling and GitHub Actions which reduced deployment breakage by 25 percent.

MaanPaa INTL PVT LTD

ML Engineer

October 2022 to May 2023

- Improved malware analysis and SIEM workflows using Python, Java, and scikit learn with structured feature pipelines, consistent labeling, and improved data ingestion which increased detection accuracy by 15 percent.
- Maintained ML operations pipelines with Docker, GitLab CI and CD, and Kubernetes with model packaging controls, version tracking, and consistent update procedures using Kubeflow which reduced deployment time.
- Unified telemetry inputs with inference systems which improved alert thresholds and reduced false positives.

Projects

Mirage: AI Powered Networking

April 2025 to May 2025

Live Demo (scroll right and click): Mirage

Tech Stack: React, D3.js, Node.js, FastAPI, Python, LangChain, ChromaDB, GraphQL, Docker, NGINX

- Built a layered circular graph interface with event propagation logic and scalable state management which improved interface load performance by 12 percent.
- Expanded ingestion workflows using Google Sheets API, n8n, and Zapier with validation and synchronization logic which reduced data inconsistency by 30 percent.

Dark Web Traffic Detection System

February 2024 to May 2024

Tech Stack: Python, XGBoost, CNN GRU, Attention Mechanism, Pandas, NumPy, Scikit learn, TensorFlow, Keras

- Built a traffic identification pipeline using XGBoost and CNN GRU with attention layers and expanded packet capture features which increased detection accuracy by 35 percent and reduced false classifications across network conditions.
- Added structured preprocessing logic for packet standardization to improve model consistency.

Education

Florida Atlantic University

Master of Computer Science, Minor in AI, AI lead for Google Development Student club

Boca Raton, FL

2024 to 2025

Certifications

AWS AI
Practitioner

Microsoft AI
Fundamentals

Cisco DevNet
Associate

AWS Cloud
Practitioner

Google Data
Analytics

Skills

- Programming:** Python, Java, JavaScript, TypeScript, SQL, C++, Bash, HTML, CSS, MLflow
- Frameworks:** TensorFlow, FastAPI, PyTorch, Scikit learn, LangChain, Hugging Face, AWS, Azure
- Data Analysis:** Redis, Prometheus, Grafana, Pandas, NumPy, Matplotlib, Seaborn, Apache Spark, vector databases
- Developer Tools :** Terraform, GraphQL, REST APIs, Git, GitHub Actions, Jira, Pinecone, Docker, Kubernetes